Week 12: November 16-November 20, 2026

الأسبوع #: من - إلى

Subject/الموضوع: Science **Unit Title عنوان الوحدة:** Understanding Matter **Grade الصف: 7**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Energy Changes in Chemical Reactions	Identify how energy is absorbed or released during chemical reactions and explain temperature changes.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Class Teams Week 12 Study Materials
Lesson 2 Energy Changes in Chemical Reactions	Analyse real-life applications of energy changes and use scientific evidence to explain how chemical reactions affect their surroundings.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.1 Activate prior knowledge using organizers, links, and pre-teaching. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Class Teams Week 12 Study Materials
Lesson 3 Properties of Matter: Mass, Volume, and Density (MUST & SHOULD)	Describe the physical properties of matter and calculate density using mass and volume.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 3.1 Activate prior knowledge using organizers, links, and pre-teaching.	Class Teams Week 12 Study Materials
Lesson 4 Properties of Matter: Density and Data Analysis (COULD & Challenge)	Analyse mass, volume, and density data to identify substances and justify scientific conclusions.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 3.1 Activate prior knowledge using organizers, links, and pre-teaching.	Class Teams Week 12 Study Materials

Homework **الواجبات**	Dear Parents, As we approach the end-of-term exams, please encourage your daughters to **begin studying now rather than waiting until the last minute**. The **Essential Reading materials**, which cover the content being tested, are available in the **Study Materials folder on MS Teams**. Regular review and practice will help build their confidence, strengthen their understanding, and ensure they are well prepared. Thank you for partnering with us to support your child's learning and success.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	One or part of a lesson maybe used for GROW Day

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	Inspire Science Digital Interactives, PPT
Cross-curricular links الروابط بين المواد الدراسية	**ELA** – Scientific vocabulary and evidence-based explanations using models and observations.
External Assessment Integration دمج التقييمات الخارجية	MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking – Critical Thinking:** Analyze scientific evidence, temperature changes, and relationships between mass, volume, and density to explain chemical reactions, compare substances, and justify conclusions using scientific reasoning. **Research – Information Literacy:** Gather and interpret scientific information from diagrams, measurements, data tables, and investigations to explain energy transfer, calculate density, and identify substances based on their physical properties.
Standards المعايير	**1. MS-PS1-2:** Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred. **2. MS-PS1-4:** Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can scientists use measurements and experimental evidence to understand the properties and changes of matter?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1:** **Must (DOK 1):** Identify endothermic and exothermic reactions and recall how energy changes when bonds break or form**.** **Should (DOK 2):** Explain how temperature changes indicate energy transfer between a reaction and its surroundings. **Could (DOK 3):** Analyze evidence to explain why chemical reactions make their surroundings warmer or cooler. **Lesson 2:** **Must (DOK 1):** Identify whether familiar chemical reactions absorb or release energy. **Should (DOK 2):** Explain how released thermal energy affects moving air and surrounding objects.

	Could (DOK 3–4): Construct scientific explanations using Claim–Evidence–Reasoning (CER) and evaluate ways to improve energy-related designs. **Lesson 3:** **Must (DOK 1):** Define mass, volume, and density and identify their measurement units. **Should (DOK 2):** Calculate density using mass and volume and determine the volume of irregular objects using water displacement. **Could (DOK 3):** Compare the densities of different materials and explain how density helps identify substances. **Lesson 4:** **Must (DOK 1):** Recall the relationship between mass, volume, and density. **Should (DOK 2):** Calculate and compare densities using experimental measurements and data tables. **Could (DOK 3–4):** Analyze experimental evidence, identify unknown substances, and justify conclusions using Claim–Evidence–Reasoning.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث
	المجال **Domain – المجال**: Citizenship **Subdomain – المجال الفرعي**: Conservation **Area – المحور:** Understanding how energy changes in chemical reactions and the physical properties of matter (mass, volume, and density) contribute to scientific innovation, sustainable engineering, healthcare, and technological development in the UAE.
	Implementation: Students appreciate the UAE's commitment to scientific innovation and sustainable development by exploring energy transfer during chemical reactions and investigating how mass, volume, and density help identify and select materials. Through scientific reasoning and data analysis, students recognize the importance of these concepts in healthcare, engineering, and environmental sustainability.